

8.4.25

EE25BTECH11022 - sankeerthan

Question: The equation of directrix of parabola $y^2 + 4y + 4x + 2 = 0$

- 1) $x = -1$
- 2) $x = \frac{-3}{2}$
- 3) $x = 1$
- 4) $x = \frac{3}{2}$

Solution: Given the parabola equation:

$$y^2 + 4y + 4x + 2 = 0 \quad (1)$$

The equation of a conic with directrix $\mathbf{n}^\top \mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, f = 2 \quad (2)$$

Find vertex $\mathbf{c}$ from

$$\mathbf{V} \mathbf{c} = -\mathbf{u} \quad (3)$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c_x \\ c_y \end{pmatrix} = -\begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad (4)$$

$$\begin{pmatrix} 0 \\ c_y \end{pmatrix} = -\begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad (5)$$

$$c_y = -2 \quad (6)$$

Put $c_y = -2$ into parabola to find c_x :

$$(-2)^2 + 4(-2) + 4c_x + 2 = 0 \Rightarrow c_x = \frac{1}{2} \mathbf{c} = \begin{pmatrix} \frac{1}{2} \\ -2 \end{pmatrix} \quad (7)$$

Eigenvector for zero eigenvalue of $\mathbf{V}$:

$$\mathbf{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} p = \frac{\|\mathbf{u}\|^2 - f}{4 \mathbf{u}^\top \mathbf{a}} = \frac{4 + 4 - 2}{8} = \frac{6}{8} = \frac{3}{4} \quad (8)$$

Equation of directrix (line perpendicular to $\mathbf{a}$, distance p from vertex):

$$\mathbf{n}^\top \mathbf{x} = \mathbf{n}^\top \mathbf{c} + p, \text{ where } \mathbf{n} = \mathbf{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mathbf{x} = \frac{1}{2} + \frac{3}{4} = \frac{5}{4} \quad (9)$$

Therefore, the equation of the directrix is approximately: $\mathbf{x} = \frac{3}{2}$

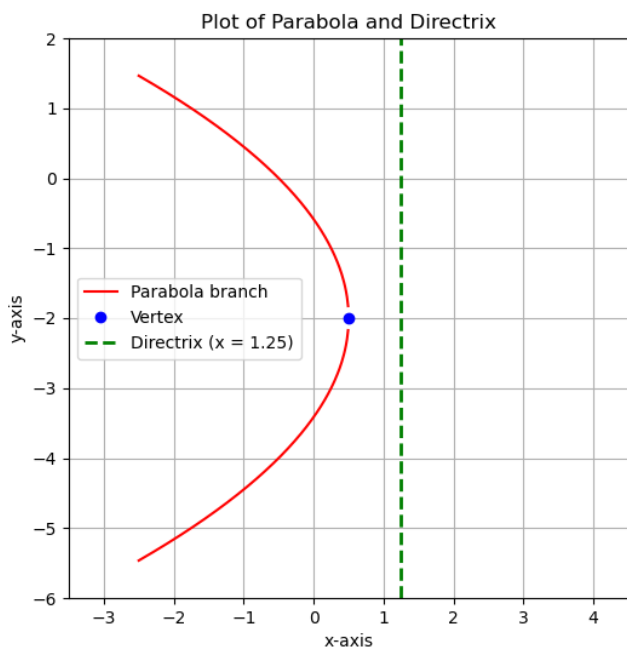


Fig. 4.1